

Data-Driven Insights from NYC Yellow Taxi Trips (2022)

SQL analysis and revenue optimisation insights

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Executive summary

Using the 2022 NYC Yellow Taxi Trips dataset in BigQuery, I answered eight business questions focused on demand concentration, fare-per-mile anomalies, and Staten Island revenue and tipping seasonality.

Key results:

- Citywide demand is highly concentrated: JFK Airport is the largest pickup hub, and Upper East Side North and South and Midtown Centre dominate everyday trips.
- The "highest fare per mile" metric is distorted around airports and geographic boundaries (for example Newark Airport), so it should be treated as an anomaly detector rather than "best pricing".
- For Staten Island pickups, Manhattan is the highest-value destination borough by total revenue, and summer (July to September) is the strongest tipping period in the available data.

Business question

How can a leader responsible for optimising revenue per taxi journey use trip-level evidence to prioritise high-value corridors, improve driver allocation, and target the periods and areas where revenue per trip is strongest?

Dataset and tools

- Dataset: `bigquery-public-data.new_york_taxi_trips.tlc_yellow_trips_2022`
- Lookup table for zones and boroughs: `bigquery-public-data.new_york_taxi_trips.taxi_zone_geom`
- Tooling: Google BigQuery (SQL) for extraction, joins, and aggregation; PowerPoint and Gamma for charts and presentation output.

Data quality and cleaning approach

I avoided a single global cleaning rule because each question requires different validity constraints (counts versus averages versus totals). Instead, I used question-specific filters that protect each metric.

Key checks:

- December 2022 contains only 57 rows, so it was excluded from time-sensitive questions (quarterly and monthly trends) to avoid misleading seasonality conclusions.
- Outliers were removed when computing distance-based or revenue totals (typical caps used: `trip_distance > 0 and <= 200 miles`; `fare_amount > 0 and <= $500`; `total_amount > 0 and <= $600`; negative tips removed; NULL tips kept as expected cash behaviour).
- Zone join integrity: I used INNER JOIN when zone names were required in the output; LEFT JOIN only when zone labels were optional. This avoids reporting "unknown" zones as business insights.
- Duplicate check: only 1 exact duplicate row was detected in approximately 36 million trips, so duplicates have negligible impact.

Best options to handle incomplete December data (if a complete time series is required):

- Option 1 - 10-year seasonal ratio (recommended): compute the December-to-November ratio for the last 10 years (for trips, tips, or revenue), take the median ratio (robust to unusual years), and apply it

to November 2022. Run separately by borough or zone to preserve geography.

- Option 2 - Recent-year December profile with trend adjustment: estimate December using the average (or weighted average) of the last 3 to 5 Decembers by borough or zone, then adjust using the change observed in January to November 2022 versus prior years so the level matches 2022 conditions.
- Option 3 - Time-series or regression forecast (most rigorous): train a model on multiple years of daily or monthly aggregates (month, day-of-week, holiday flags, year trend, borough or zone effects) and predict December 2022 with a prediction interval. Use this when operational accuracy matters and enough history is available.

Findings (Q1–Q7)

Q1 - Top 5 pickups and drop-offs

Pickups: JFK Airport; Upper East Side South; Upper East Side North; Midtown Centre; Penn Station/Madison Sq West.

Drop-offs: Upper East Side North; Upper East Side South; Midtown Centre; Penn Station/Madison Sq West; Times Square and Theatre District.

Q2 - Top 3 zones with highest average fare per mile (anomalies)

Newark Airport shows an extreme value because only mileage inside NYC limits is recorded while the full fare (including fixed charges) applies; Kew Gardens and South Ozone Park reflect very short airport-related trips where minimum fares and surcharges inflate the ratio.

Q3 - Staten Island pickups: revenue by destination borough and top zones

Destination borough revenue (total_amount): Manhattan is highest (\$64,179), followed by Brooklyn (\$39,128) and Staten Island (\$38,786).

Top drop-off zones by revenue include Sheepshead Bay (Brooklyn), Arden Heights (Staten Island), and East Harlem South (Manhattan), showing revenue is concentrated in specific corridors rather than evenly spread.

Q4 - Staten Island tips by quarter (UK quarters as instructed)

Q1 (Apr to Jun): \$981

Q2 (Jul to Sep): \$1,045 (highest in available data)

Q3 (Oct to Dec): \$828 - note: December is incomplete, so Q3 is effectively Oct to Nov for this dataset

Q4 (Jan to Mar): \$807 - for this task, Q4 is treated as Jan to Mar 2022 (per instructions). In a real-world year-end quarterly view, Q4 would typically refer to Jan to Mar 2023 to complete the full rolling year.

Q5 - Zones with > \$100 in tips during Q2

Heartland Village and Todt Hill (\$136), Bloomfield and Emerson Hill (\$112), South Beach and Dongan Hills (\$104).

Q6 - Tip efficiency (tips per trip) in Q2 (min 10 trips)

Heartland Village and Todt Hill is the clear leader at about \$3.02 per trip (45 trips), far above Bloomfield and Emerson Hill (about \$0.34 per trip; 327 trips). South Beach and Dongan Hills was excluded here due to fewer than 10 trips.

Q7 - Monthly trip count trend (Staten Island pickups)

Trips rise from winter lows and stay high between June and October, peaking in August. December is excluded due to incompleteness.

Top 3 insights for revenue optimisation (Q8)

1) Prioritise Staten Island to Manhattan corridors (fare plus tip)

Manhattan-bound journeys generate the highest revenue for Staten Island pickups. Operationally, this implies ensuring enough driver availability in the Staten Island pickup zones and time windows that produce these long, high-value trips.

2) Use seasonality to allocate supply (and re-check Q3 when full data is available)

The data shows stronger tipping and higher demand through summer, with July and August standing out. Driver scheduling and fleet allocation should be optimised for these months. Because December is incomplete, Q3 should be re-validated if complete data becomes available (or imputed with a clearly stated method).

3) Target high-efficiency tipping zones

Some Staten Island pickup zones generate disproportionately high tips per trip (not just high totals). Heartland Village and Todt Hill is the clearest example, delivering materially higher revenue per journey. Focused service and availability improvements in these zones can lift revenue per trip faster than spreading resources evenly across the borough.

Limitations and next steps

- Staten Island has relatively low Yellow Taxi volume compared with Manhattan and Queens; results should be interpreted as directional for prioritisation and validated with additional years and (ideally) Green Taxi and FHV data for borough-level coverage.
- The quarter analysis is affected by incomplete December data. A complete 12-month series is required before making firm Q2 versus Q3 comparisons.
- Next steps: add time-of-day and day-of-week analysis for the Staten Island to Manhattan corridor; compare revenue per trip by payment type (card versus cash); and examine whether tip-per-trip varies by journey purpose proxies (airport versus local) and route distance bands.